

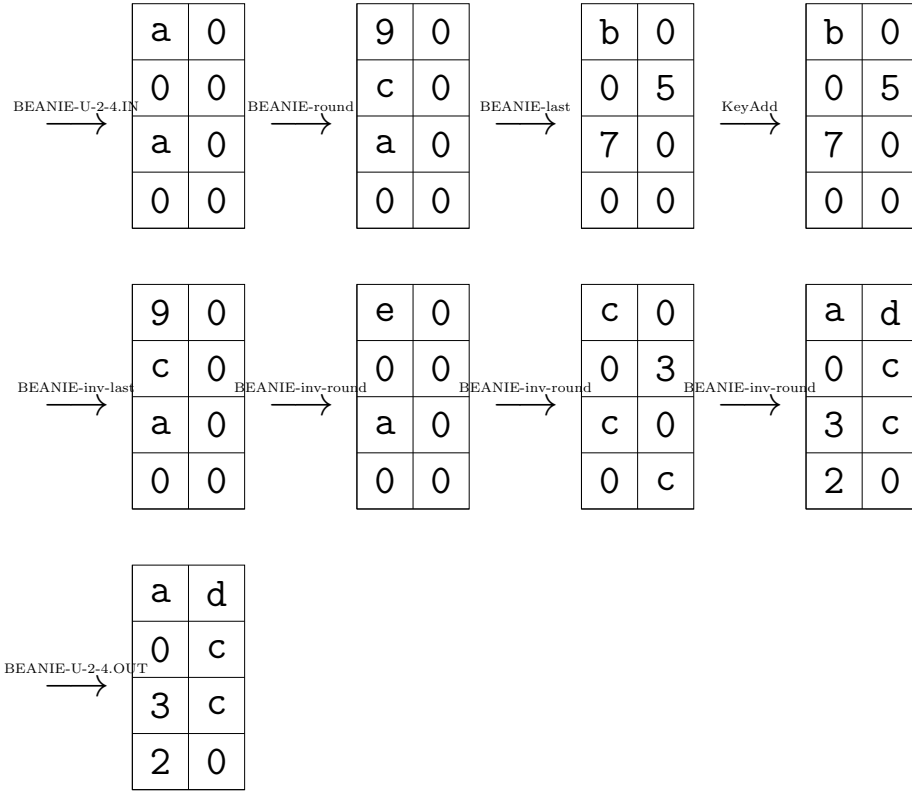
Bitwise differential trail through **BEANIE-U-2-4** found by SAT

CiVerLy

July 16, 2026

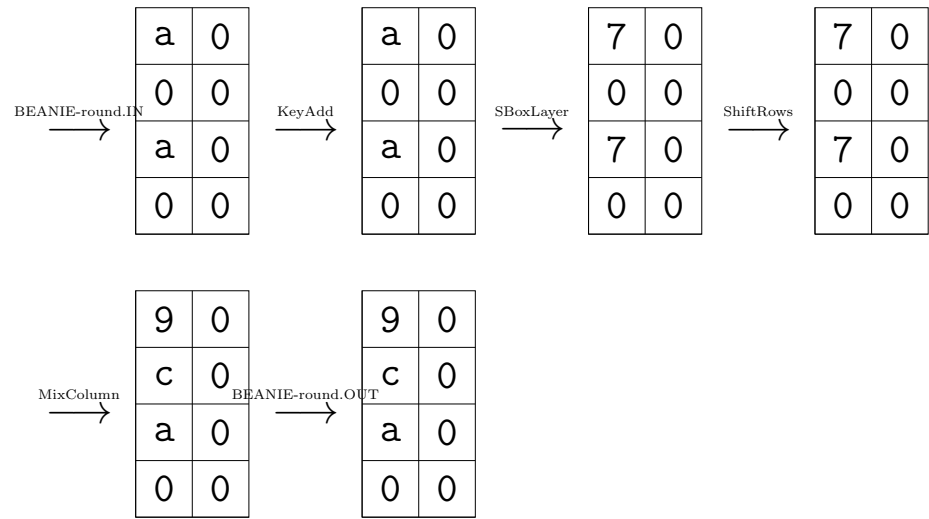
1 BEANIE-U-2-4

Maximal differential probability: 2^{-40}



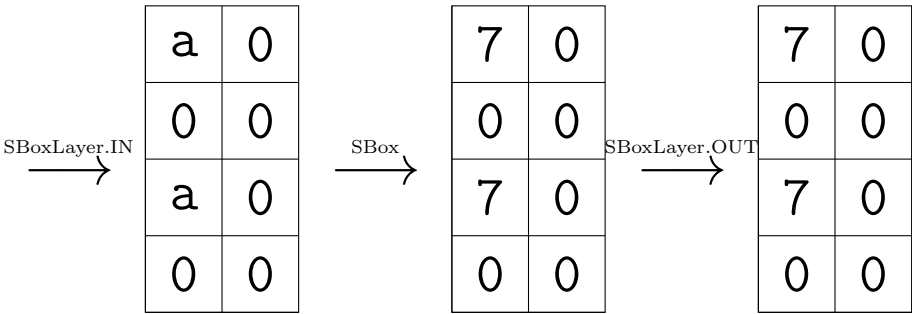
2 BEANIE-round

Maximal differential probability: $2^{-4.0}$



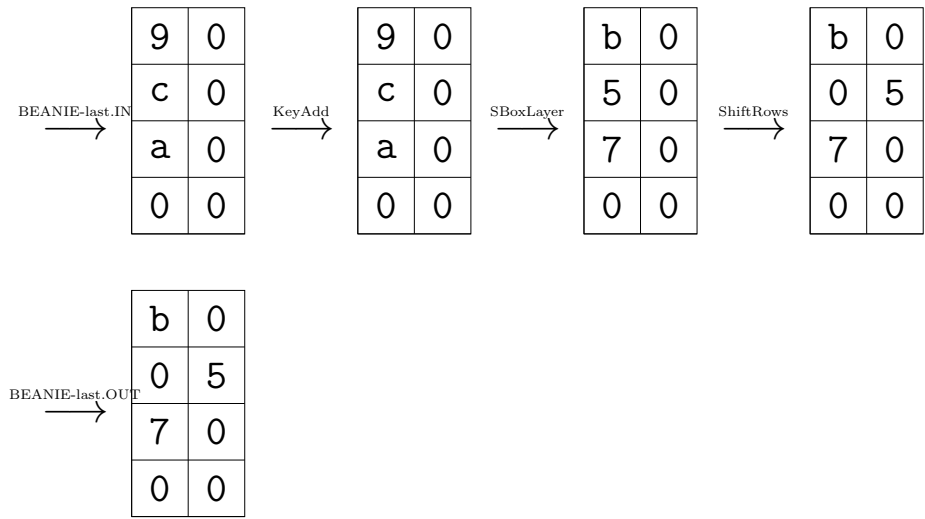
3 SBoxLayer

Maximal differential probability: $2^{-4.0}$



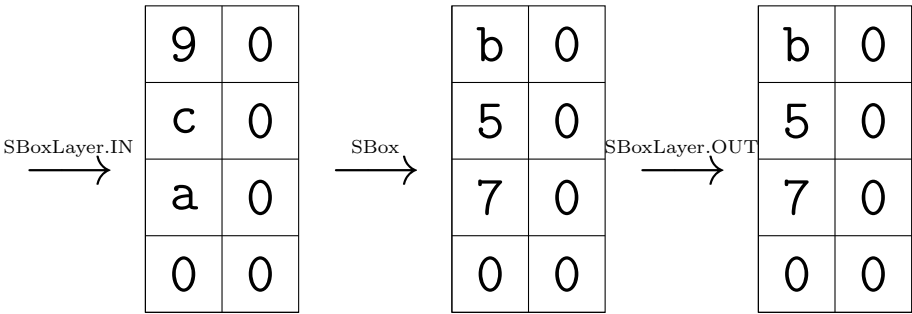
4 BEANIE-last

Maximal differential probability: $2^{-6.0}$

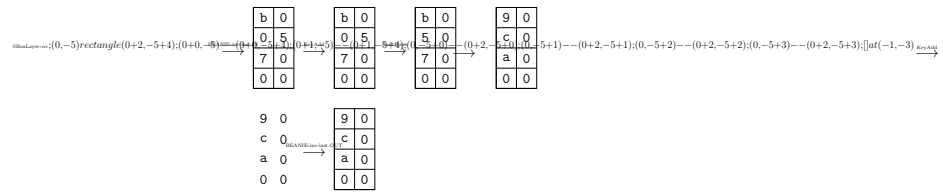


5 SBoxLayer

Maximal differential probability: $2^{-6.0}$



6 BEANIE-inv-last

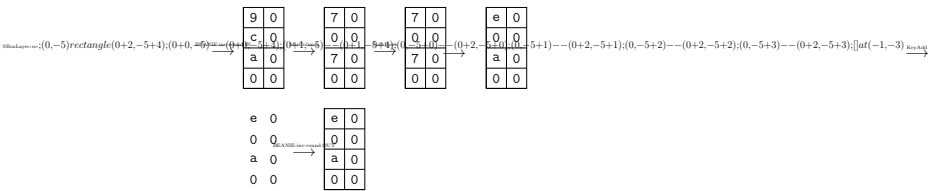
Maximal differential probability: $2^{-6.0}$ 

7 SBoxLayer_inv

Maximal differential probability: $2^{-6.0}$

8 BEANIE-inv-round

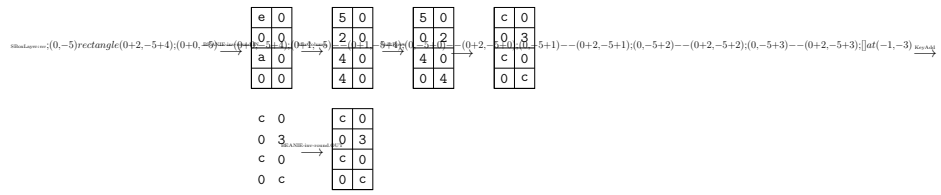
Maximal differential probability: $2^{-4.0}$



9 SBoxLayer_inv

Maximal differential probability: $2^{-4.0}$

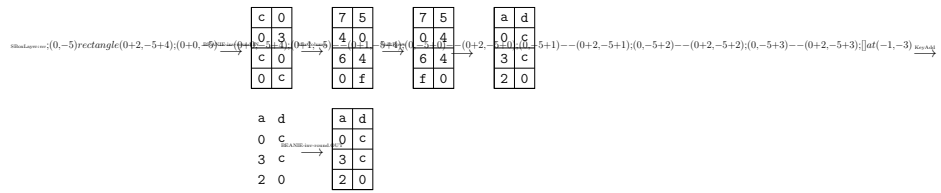
10 BEANIE-inv-round

Maximal differential probability: $2^{-8.0}$ 

11 SBoxLayer_inv

Maximal differential probability: $2^{-8.0}$

12 BEANIE-inv-round

Maximal differential probability: $2^{-12.0}$ 

13 SBoxLayer_inv

Maximal differential probability: $2^{-12.0}$